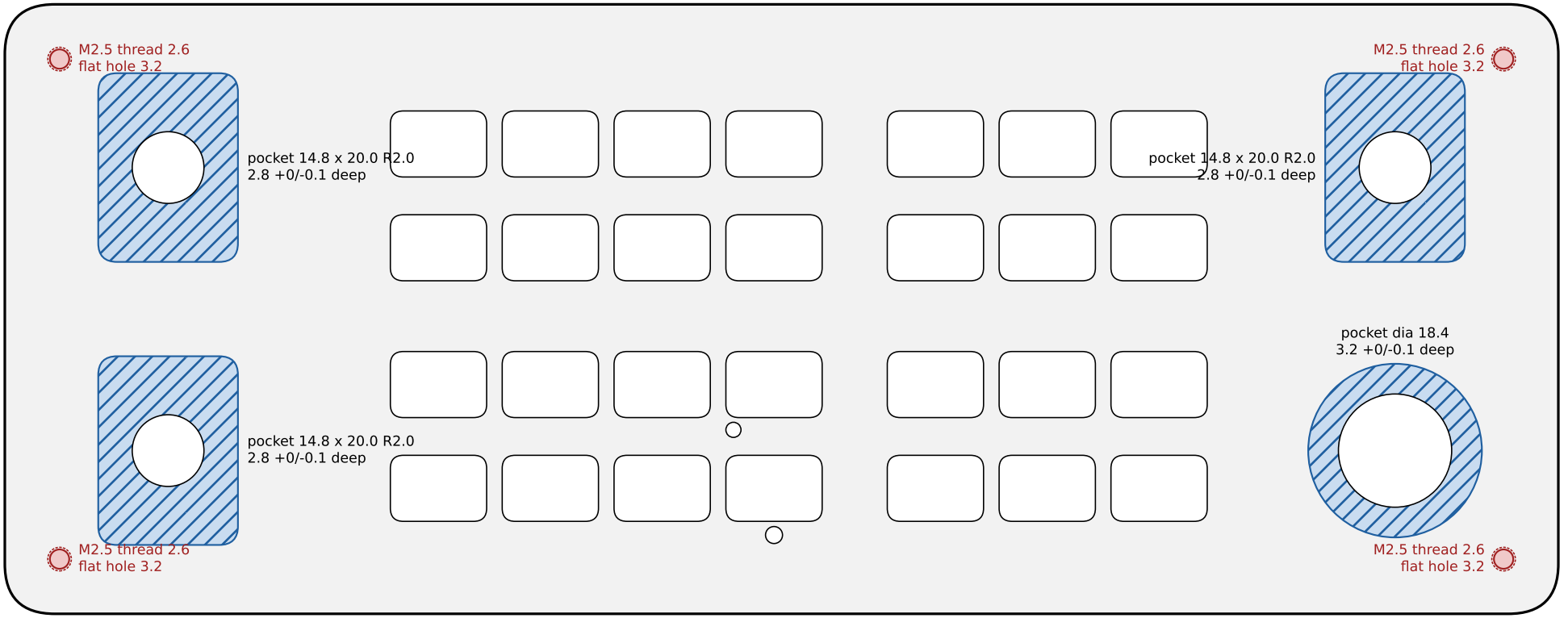
Avionics Panel rev 1 front plate, seen from the BACK (machined side)



Features

key hole	10.2 x 7.0 R1.4	through	x28
encoder hole	dia 7.6	through	x3
encoder pocket	14.8 x 20.0 R2.0	2.8 +0/-0.1 deep from back	x3
RANGE hole	dia 12.0	through	x1
RANGE pocket	dia 18.4	3.2 +0/-0.1 deep from back	x1
SHIFT light hole	dia 1.8	through	x1
UNIT light hole	dia 1.6	through	x1
threaded hole	M2.5, thread milled	full thread 2.6 min, flat hole 3.2 +0/-0.1, from back	x4

Notes

- Material 6061 aluminium, 4.0 +-0.1 mm. Outline 164.6 x 64.6 mm, corner R5.3. Volume 31.0 cm3. Units mm, general tolerance ISO 2768-m.
- Front face (+Z in the STEP) is flat: no pockets, chamfers or fastener holes on the front.
- Positions of the 4 threads and all through holes: +-0.1 to each other (the key caps and the screws must line up with the board).
- 4 x M2.5 blind threads from the back, THREAD MILLED in a flat-bottom hole (end mill, no drill point): hole 3.2 +0/-0.1 mm deep, full thread from 0.1 to at least 2.6 mm below the back face (deburr only, no countersink). Keep at least 0.7 mm of material to the front face under every hole and pocket (depths +0, never deeper): nothing may break through or mark the front face.
- Threads: mask them before anodising, or chase them after anodising (the coating makes them tight).
- Finish: black anodised, matte (bead blasted before anodising).
- Marking on the front per avionics-panel-rev1-plate.dxf, same origin (bottom-left, seen from the front): layer LASER = laser marking, layer UV_ORANGE = UV print orange. Layer CUT = outline and through holes for reference.
- Deburr all edges, edge break 0.2 on the front outline and hole edges.